

AZTIBASE NETWORK

The Chain Where AI Agents Live

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aztibase.com | [@aztibase](https://twitter.com/aztibase)

The Problem

AI agents need a blockchain that understands them. Current chains treat AI as an afterthought — inference happens off-chain, verification is impossible, and 12-second block times make per-inference micropayments impractical.

Aztibase is the first Layer 1 blockchain built from the ground up for AI agent economies.

What Makes Aztibase Different

AI-Native at the Protocol Level

Every Aztibase node runs an ONNX inference engine. AI isn't a smart contract bolted on top — it's part of the node software itself.

- **Model Registry** — Register AI models on-chain with metadata, pricing, and version tracking
- **Inference Execution** — Nodes execute AI inference tasks and produce verifiable results
- **AI Sentinel** — Autonomous health monitor using machine learning, detecting anomalies and taking corrective action
- **Agent Identity** — On-chain accounts for AI agents with capability policies and spending limits

Dual Virtual Machine

Aztibase runs both EVM and WASM smart contracts side by side, with a cross-VM bridge:

- **EVM (revm)** — Deploy Solidity contracts, compatible with Ethereum tooling
- **WASM (wasmtime)** — Deploy Rust, C, or AssemblyScript for maximum performance
- **Cross-VM Bridge** — WASM contracts call EVM contracts and vice versa

DAG-Based Consensus (Synaptic Consensus)

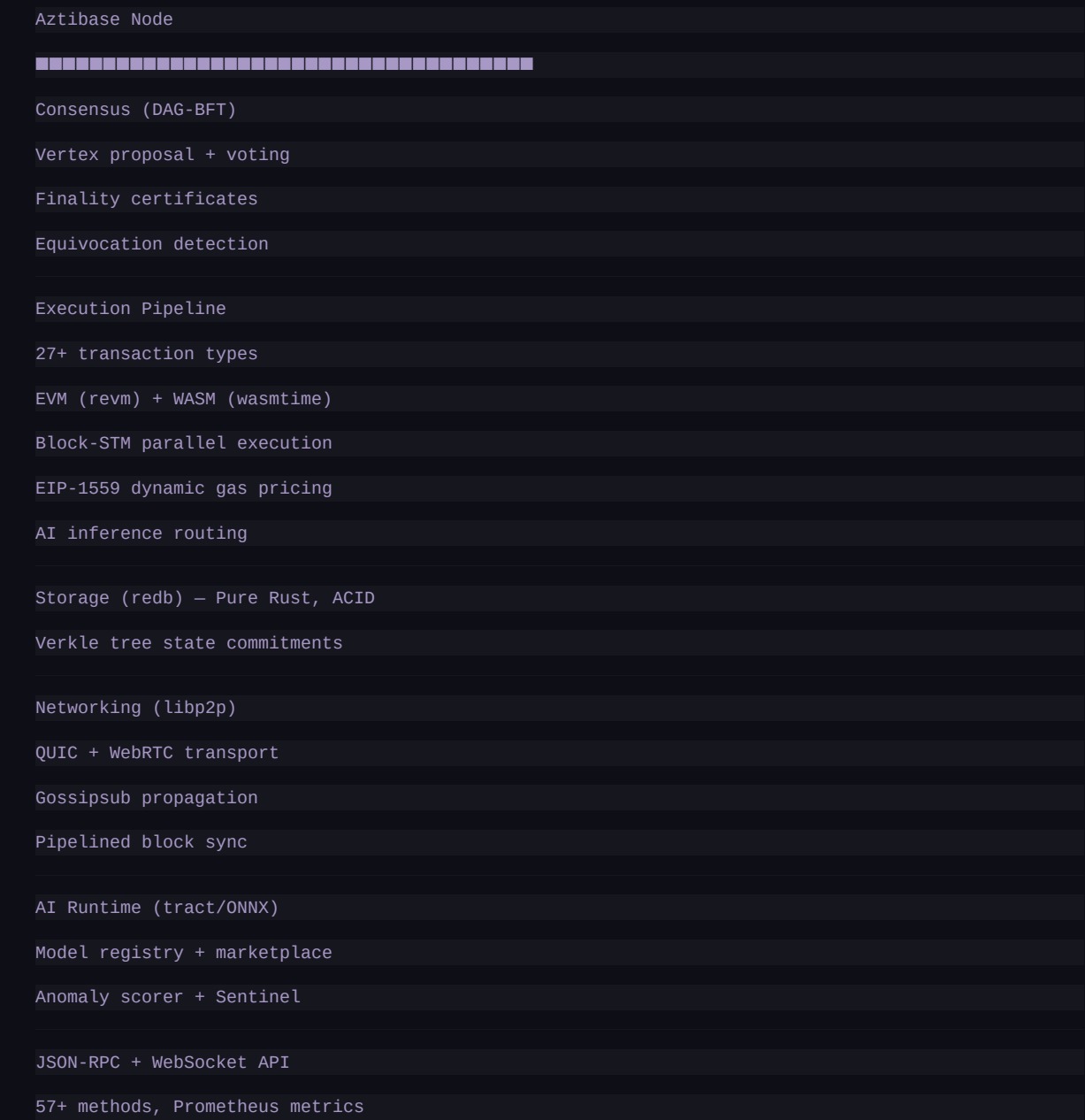
Unlike linear blockchains, Aztibase uses a Directed Acyclic Graph where multiple blocks are proposed simultaneously:

- **400ms block time** (vs 12s Ethereum)
- **Sub-1.6s finality** (4 consensus rounds)
- **No leader bottleneck** — all validators propose concurrently
- **Byzantine fault tolerant** — tolerates up to 1/3 malicious validators

Server Independence

- Pure P2P via libp2p (Kademlia DHT, gossipsub, QUIC + WebRTC)
- Every node is its own RPC endpoint
- No bootstrap server dependency after initial peer discovery
- Browser nodes via WASM + WebRTC (no server relay)

Architecture



What's Built and Running

Component	Status	Details
DAG Consensus	Live	Sub-second finality on public testnet
EVM Contracts	Live	Deploy, call, read via aztb_call/eth_call
WASM Contracts	Live	Full deploy/call with fuel metering
Cross-VM Bridge	Live	Bidirectional EVM-WASM calls

Component	Status	Details
Block-STM	Live	Concurrent transfer execution
AI Sentinel	Live	3-tier autonomous action engine
Dynamic Gas	Live	EIP-1559 adaptive base fee
DEX (AMM)	Live	WASZTB/tUSDC pair with liquidity
Oracle	Live	CoinGecko price feed, on-chain updates
Block Sync	Live	5x pipelined catch-up
TypeScript SDK	Published	@aztibase/sdk, 21 tests
Chrome Wallet	Published	Extension with staking UI
Indexer + Explorer	Live	SQLite-backed, REST API

Tokenomics

AZTB Token

Parameter	Value
Total Supply	1,000,000,000 AZTB
Genesis Mint	400,000,000 (40%)
Emission Pool	600,000,000 (60%) over ~10 years
Block Time	400ms
Consensus	DAG-BFT (Synaptic Consensus)

Genesis Allocation (400M)

Category	Amount	Vesting
Protocol Treasury	100M (25%)	Immediate, governance-controlled
Ecosystem Development	80M (20%)	6-month cliff, 4-year linear
Core Team	60M (15%)	12-month cliff, 4-year linear
Foundation Reserve	40M (10%)	2-year lock, 3-year linear
Community Airdrop	40M (10%)	Immediate
Validator Bootstrap	40M (10%)	6-month linear
AI Ecosystem Fund	20M (5%)	6-month cliff, 3-year linear
Liquidity Provision	20M (5%)	Immediate

Emission Schedule

Validator rewards, AI compute incentives, and treasury funding come from the 600M emission pool:

Years	Annual Emission	Cumulative
1-2	120M/year	240M
3-4	60M/year	360M
5-6	30M/year	420M
7-8	15M/year	450M
9-10	7.5M/year	465M
11+	3.75M/year (tail)	Approaches 600M

Emission Distribution

Recipient	Share
Validator Rewards	70%
AI Compute (PoUW)	15%
Protocol Treasury	10%
Staking Insurance	5%

Staking

Parameter	Value
Minimum Stake	50,000 AZTB (governance range: 10K-500K)
Maximum Stake	50,000,000 AZTB (5% of supply)
Unbonding Period	21 days
Target APY	3-12%
Equivocation Slash	10%
Downtime Slash	0.5%

AI Integration

Three-Layer AI Stack

Layer 1 — Anomaly Detection (per-transaction)

Every transaction is scored for anomalous behavior using an ONNX autoencoder model running inside the node. High scores flag potential attacks or unusual patterns.

Layer 2 — Sentinel Health Monitor (per-epoch)

The AI Sentinel aggregates network metrics (commit latency, TPS, peer count, equivocations) and produces a chain health score. It operates in three tiers:

- Tier 1: Observe and report
- Tier 2: ONNX model-based scoring with confidence levels
- Tier 3: Autonomous action (throttle gas, alert validators, trigger emergency responses)

Layer 3 — Compute Marketplace

Model creators register models on-chain. Users post inference tasks with reward amounts. Compute providers execute tasks, submit attestations, and receive payment. The chain verifies results.

Why This Matters

AI agents generating autonomous transactions need a chain that can:

1. Verify inference results (not just trust an oracle)
2. Settle micropayments per-inference (400ms finality enables this)
3. Manage agent identity and spending policies on-chain
4. Detect and respond to anomalous agent behavior automatically

No other L1 has this stack built into the protocol layer.

Technology

Component	Choice	Why
Language	Rust	No GC, memory safety, WASM target
Async Runtime	Tokio	Production-grade async I/O
P2P	libp2p	NAT traversal, QUIC, WebRTC

Component	Choice	Why
EVM	revm	Production Ethereum execution engine
WASM VM	wasmtime	Fastest WASM runtime
Storage	redb	Pure Rust, ACID, no C deps
State	Verkle trees	Constant-size proofs
Hashing	BLAKE3	Fastest cryptographic hash
Signatures	Ed25519	Industry standard
AI Runtime	tract (ONNX)	Pure Rust inference, no Python
Serialization	postcard	Compact binary, no-std

9 Rust crates in monorepo. 1,040+ tests, zero flaky. 34 Architecture Decision Records. MIT / Apache-2.0 dual license.

Roadmap

Completed

- ✓ Layer 1 blockchain (consensus, execution, networking, storage)
- ✓ Public testnet with monitoring (Grafana + VictoriaMetrics)
- ✓ Dual VM (EVM + WASM) with cross-VM bridge
- ✓ AI Sentinel with ONNX inference
- ✓ On-chain governance (proposals, voting, parameter changes)
- ✓ L2 bridge primitives (RegisterL2, AnchorL2State, BridgeDeposit, BridgeWithdraw)
- ✓ AMM DEX with liquidity pools
- ✓ Oracle with live price feeds
- ✓ TypeScript SDK, Chrome wallet extension
- ✓ Block explorer and indexer

Next

- Security audit
- Mainnet launch with 100+ validators
- AI Model Marketplace (register, price, and trade AI models)
- Agent Identity Registry (on-chain DID for AI agents)
- Content Provenance Registry (C2PA-style proof of AI-generated content)
- Name Service (.aztb domains)
- Community Task Chain (L2 for AI compute tasks)

How to Participate

Run a Validator

Download the binary from GitHub, stake AZTB, and start earning rewards. Minimum hardware: 2 vCPU, 4 GB RAM, 50 GB SSD (~\$4/month on Hetzner).

Build on Aztibase

Deploy EVM (Solidity) or WASM (Rust) contracts. Full TypeScript SDK available.

Join the Community

- Website: aztibase.com
- Twitter: [@aztibase](https://twitter.com/aztibase)
- GitHub: github.com/aztibase/aztibase-node

Team

Built by a solo engineer in South Africa. Every line of code — consensus, networking, execution, AI, wallet, explorer, SDK — written from scratch in Rust.

Aztibase (Pty) Ltd is registered with the South African CIPC.

Aztibase Network — The Chain Where AI Agents Live